



Cloudflare Radar

2024 Year in Review

Internet patterns and trends observed through Cloudflare's global network

2024

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

YoY Global Internet Traffic Growth

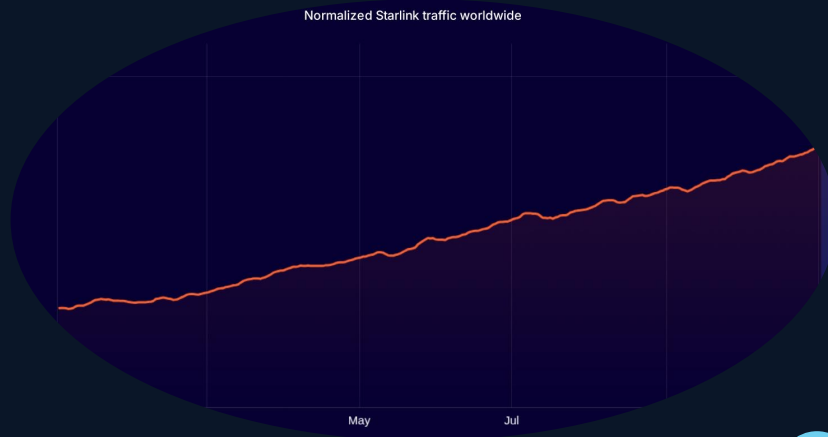
3.3x

Starlink Traffic Surge in 2024

Traffic Growth Drivers

Growth driven by increasing reliance on Internet for communication, commerce, entertainment, and transportation

Trend line starts mid-January to normalize for post-holiday activity patterns



Normalized Starlink traffic worldwide — 2024

Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

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Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

2% → 13%

Post-Quantum
TLS 1.3 Traffic

21%

HTTP/3
Adoption Rate

Go #1

API Client Language
(~12% of traffic)

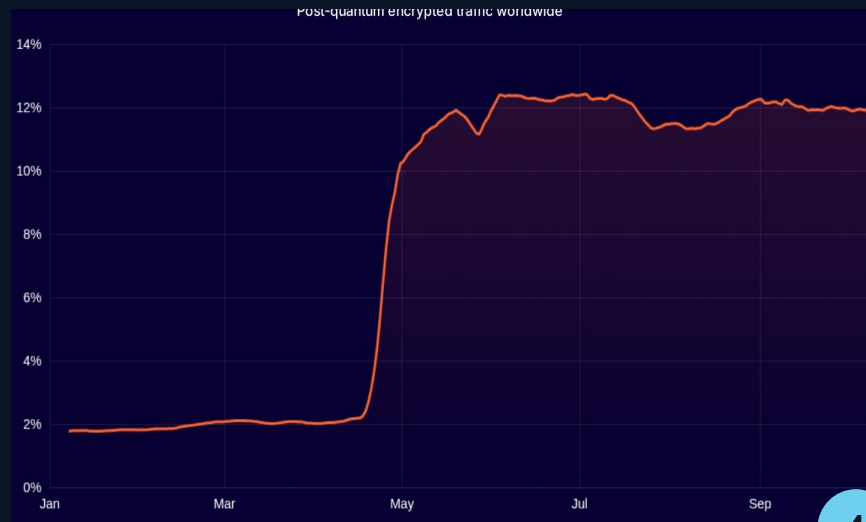
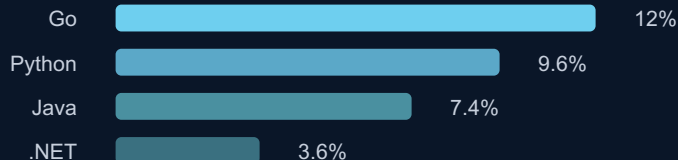
Post-Quantum Adoption

Chrome 124 enabled PQ key agreement by default (Apr 17, 2024)

Mozilla Firefox rolling out PQ encryption by default

Apple Safari began initial PQ testing

API Client Languages



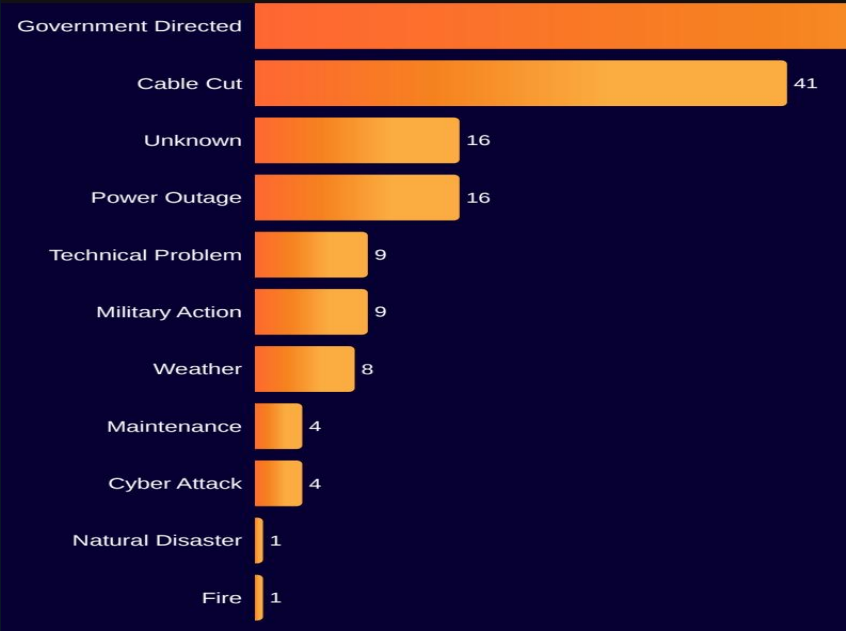
Post-quantum encrypted traffic adoption after Chrome 124 release

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

major Internet disruptions
observed globally in 2024

Cause	Count	
Government Directed	116+	
Cable Cut	41	
Unknown	16	
Power Outage	16	
Technical Problem	9	
Military Action	9	
Weather	8	
Maintenance	4	
Cyber Attack	4	
Natural Disaster	1	
Fire	1	



28%

IPv6 Adoption

41%

Mobile Traffic Share

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

All Traffic
Mitigated

4.3%

Emails Flagged
as Malicious

68%

Bot Traffic from
Top 10 Countries

Attack Landscape

Most Attacked Sector: Gambling & Games

Log4j (CVE-2021-44228) — still actively exploited 2+ years after initial disclosure, underscoring persistent nature of widely known vulnerabilities

RPKI valid routes increased **6.4%**, improving Internet routing security

Email Security Threats

43%

Malicious emails containing deceptive links

.bar

Top malicious/spam TLD origin

4.3%

Total emails flagged as malicious

Five Takeaways Shaping the 2025 Internet Outlook

1

Traffic Growth Trajectory

Global Internet traffic grew 17% YoY; Starlink surged 3.3x expanding satellite connectivity to underserved regions.

2

Post-Quantum Momentum

Post-quantum encryption leapt from 2% to 13% of TLS 1.3 traffic after Chrome 124 default enablement in April 2024.

3

Connectivity Fragility

Government-directed shutdowns led 225 major disruptions, followed by cable cuts (41) and power outages (16).

4

Persistent Security Threats

6.5% of traffic mitigated; Gambling/Games most targeted sector; Log4j exploitation persists years after disclosure.

5

Evolving Developer Ecosystem

Go became the top API client language at ~12% of automated API traffic, reflecting shifts in how developers build for the Internet.